

University of Verona

Department of Computer Science

MSc in Computer Engineering

Transportation and Logistics Ontology

OWL 2.0 Project Report

Course	Knowledge Representation
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1. Introduction

This report presents the design and implementation of a Transportation and Logistics Ontology developed as part of the Knowledge Representation course exam at the University of Verona. The ontology is specified in OWL 2.0 using the RDF/XML serialization format and was modelled and validated using Protégé 5.x.

The Transportation and Logistics domain was selected due to its inherent complexity and real-world relevance. Modern logistics networks involve a rich set of interrelated concepts including vehicles, facilities, shipments, routes, personnel, tracking devices, communication protocols, and operational events. Representing these relationships formally in an ontology enables automated reasoning, semantic querying, and knowledge sharing across heterogeneous systems.

The ontology is publicly available on GitHub at:

<https://github.com/abdulraheemrauf/TransportationLogistics-Ontology>

2. Domain Description

The ontology models a European transportation and logistics network called EuroLog. The domain encompasses the following key areas:

- Physical infrastructure: warehouses, distribution centers, ports, customs checkpoints, and last-mile delivery hubs across multiple geographic regions.
- Transport units: ground vehicles (trucks, vans, refrigerated vehicles, train wagons), air vehicles (cargo planes, delivery drones), and maritime vessels (container ships, barges).
- Tracking and monitoring: GPS trackers, RFID scanners, and temperature loggers attached to vehicles and shipments.
- Shipments: general parcels, bulk cargo, and hazardous goods transported between facilities.
- Routes: domestic and international planned paths between origin and destination points.
- Personnel: logistics managers, drivers, warehouse operators, and customers.
- Logistics events: delivery events, delay events, and customs clearance events recorded during operations.
- Digital services: real-time tracking, fleet management, and warehouse management systems.
- Communication protocols: GPS/NMEA, RFID, MQTT, and EDI/EDIFACT used by tracking devices and logistics systems.

3. Class Hierarchy

The ontology defines 53 classes organized under two top-level root classes: Physical Entity and Abstract Entity. This separation cleanly distinguishes tangible real-world objects from conceptual or informational entities.

3.1 Physical Entity

Physical Entity is the root for all physical things in the logistics network. It has four direct subclasses:

- **Location** — spatial regions including LogisticsNetwork, Region, Facility (with subclasses Warehouse, DistributionCenter, Port, CustomsCheckpoint, DeliveryHub), and RoadSegment.
- **Vehicle** — transport units are divided into GroundVehicle (Truck, Van, TrainWagon, RefrigeratedVehicle), AirVehicle (CargoPlane, DeliveryDrone), and MaritimeVehicle (ContainerShip, Barge). GroundVehicle, AirVehicle, and MaritimeVehicle are declared disjoint.
- **Person** — human participants including LogisticsManager, Driver, WarehouseOperator, and Customer. Driver and Customer are declared disjoint.
- **TrackingDevice** — electronic monitoring equipment including GPSTracker, RFIDScanner, and TemperatureLogger.

3.2 Abstract Entity

Abstract Entity is the root for non-physical conceptual entities. It has four direct subclasses:

- **Shipment**—consignments of goods including Parcel, BulkCargo, and HazardousCargo.
- **Route**—planned transport paths including DomesticRoute and InternationalRoute.
- **LogisticsEvent** — recorded operational events including DelayEvent, DeliveryEvent, and CustomsClearanceEvent.
- **LogisticsService** — digital service platforms including TrackingService, FleetManagementService, and WarehouseManagementService.
- **CommunicationProtocol** — data protocols including GPSProtocol, RFIDProtocol, EDIProtocol, and MQTTProtocol.

4. Object Properties

The ontology defines 19 object properties capturing relationships between individuals. Key properties include:

Property	Domain	Range	Notes
isLocatedAt	Vehicle	Location	Current location
hasVehicle	Location	Vehicle	Inverse of isLocatedAt

isPartOf	Location	Location	Transitive
hasPart	Location	Location	Inverse + Transitive
isOperatedBy	Vehicle	Driver	Driver assignment
operates	Driver	Vehicle	Inverse of isOperatedBy
carriesShipment	Vehicle	Shipment	Active cargo
isCarriedBy	Shipment	Vehicle	Inverse
followsRoute	Vehicle	Route	Planned path
equippedWith	Vehicle	TrackingDevice	Device on vehicle
usesProtocol	TrackingDevice	CommunicationProtocol	Data protocol
generates	TrackingDevice	LogisticsEvent	Event generation
involves	LogisticsEvent	Person	Person involved
originatesAt	Shipment	Facility	Origin facility
destinedFor	Shipment	Location	Destination
ownedBy	Shipment	Customer	Shipment owner
integratedWith	LogisticsNetwork	LogisticsService	Service integration
managedBy	LogisticsNetwork	LogisticsManager	Network manager
hasAccess	Person	Facility	Access rights

5. Datatype Properties

The ontology defines 18 datatype properties for storing concrete data values. Two properties are declared as owl:FunctionalProperty, ensuring uniqueness:

- vehicleID (FunctionalProperty) — unique fleet identifier for each vehicle (xsd:string).
- shipmentID (FunctionalProperty) — unique tracking number for each shipment (xsd:string).

Other notable datatype properties include:

- isActive (xsd:boolean) — whether a vehicle is currently in service.
- fuelLevel (xsd:integer) — fuel or charge level as a percentage (0-100).
- maxPayloadKg (xsd:float) — maximum cargo weight capacity in kilograms.
- currentSpeedKmh (xsd:float) — current vehicle speed in km/h.
- temperatureReading (xsd:float) — cold-chain temperature in degrees Celsius.
- weightKg (xsd:float) — shipment weight in kilograms.
- isHazardous (xsd:boolean) — flags dangerous goods shipments.
- facilityCapacityM2 (xsd:float) — storage area in square metres.
- routeDistanceKm (xsd:float) — total planned route distance.
- eventTimestamp (xsd:dateTime) — precise timestamp of logistics events.
- personName (xsd:string) — full name of a person.

6. OWL 2 Features Used

The ontology makes use of several advanced OWL 2 constructs to increase expressivity and enable reasoning:

6.1 Disjoint Classes

Three pairs of disjoint class axioms are declared to prevent logical overlap:

- GroundVehicle, AirVehicle, and MaritimeVehicle are mutually disjoint — a vehicle cannot belong to more than one transport mode.
- Driver and Customer are disjoint — a person cannot simultaneously be a driver and a customer in this domain.

6.2 Qualified Cardinality Restriction

A qualified cardinality restriction (`owl:qualifiedCardinality = 1`) is applied to the Vehicle class on the `isLocatedAt` property, constrained to Location. This axiom states that every vehicle must be located at exactly one location at any given time, enforcing a key domain constraint.

6.3 Existential Restrictions (`someValuesFrom`)

Three existential restrictions enforce mandatory relationships:

- LogisticsNetwork must have at least one Facility as part of it (`hasPart some Facility`).
- Every Shipment must originate at at least one Facility (`originatesAt some Facility`).
- Every Shipment must be destined for at least one Location (`destinedFor some Location`).
- LogisticsManager must have access to at least one Facility (`hasAccess some Facility`).

6.4 Transitive Properties

The `isPartOf` and `hasPart` object properties are declared transitive. This means that if Warehouse `isPartOf` NorthernRegion, and NorthernRegion `isPartOf` EuroLogNet, then the reasoner can infer that Warehouse `isPartOf` EuroLogNet — enabling powerful hierarchical spatial reasoning.

6.5 Inverse Properties

Multiple inverse property pairs are declared to support bidirectional querying: `isLocatedAt` / `hasVehicle`, `isPartOf` / `hasPart`, `isOperatedBy` / `operates`, `carriesShipment` / `isCarriedBy`.

6.6 Functional Properties

vehicleID and shipmentID are declared as owl:FunctionalProperty, meaning each vehicle or shipment can have at most one identifier value, which also supports individual identity reasoning (owl:sameAs inferences).

7. Named Individuals

The ontology includes 34 named individuals providing a realistic example instance of the EuroLog transportation network:

7.1 Locations and Facilities

- EuroLogNet — the root logistics network instance.
- NorthernRegion, SouthernRegion — two geographic operational regions.
- WarehouseAmsterdam (12,000 m²), DistCenterMilan (8,500 m²), PortRotterdam (50,000 m²), CustomsBorderFrance, DeliveryHubLisbon (2,200 m²).

7.2 Vehicles

- Truck_NL_001 — Volvo FH16 long-haul truck, 24,000 kg payload, currently at 88.5 km/h on the Amsterdam-Milan route.
- Van_PT_001 — Mercedes Sprinter last-mile delivery van, operated by Driver Carlos in Lisbon.
- RefrigVehicle_IT_001 — DAF XF refrigerated truck carrying a cold-chain pharmaceutical shipment at -4.5°C.
- ContainerShip_ROT_001 — MV EuroFreight I container vessel at Port of Rotterdam.

7.3 Tracking Devices and Protocols

- GPS_Truck001, GPS_Van001, GPS_Ship001 — GPS trackers using GPS NMEA 2.3 protocol.
- TempLogger_Refrig001 — temperature logger using MQTT 3.1.1, recording -4.5°C.
- Protocol instances: GPS NMEA 2.3, RFID ISO 18000-6C, MQTT 3.1.1, EDIFACT UN/EDIFACT D.96A.

7.4 Persons

- Manager_Ahmed (LogisticsManager) — has access to Amsterdam, Milan, and Lisbon facilities.
- Driver_Carlos (Driver) — operates Van_PT_001 on the Lisbon urban delivery route.
- Operator_Jan (WarehouseOperator) — works at Amsterdam Central Warehouse.
- Customer_Maria (Customer) — owner of both active shipments.

7.5 Shipments and Routes

- Shipment_GEN_001 (Parcel, 15.5 kg) — from Amsterdam to Lisbon, owned by Maria.
- Shipment_COLD_001 (Parcel, 340 kg) — cold-chain pharmaceutical from Milan to Lisbon.
- Route_NL_to_IT (InternationalRoute, 1,285 km) — Amsterdam to Milan.
- Route_Lisbon_Local (DomesticRoute, 38.5 km) — Lisbon urban delivery circuit.

7.6 Events and Services

- DelayEvent_001 — border delay at French customs checkpoint (2026-05-12 14:20).
- DeliveryEvent_001 — successful delivery at Lisbon hub (2026-05-13 10:05).
- CustomsEvent_001 — customs clearance at French border (2026-05-12 17:45).
- Services: EuroLog Real-Time Tracking Platform, EuroLog Fleet Manager, EuroLog WMS.

8. Reasoner Validation

The ontology was validated using the Hermit 1.4 reasoner integrated within Protégé 5.x. The reasoner completed successfully with no inconsistencies detected, confirming that:

- All class axioms, property restrictions, and individual assertions are logically consistent.
- No individual is classified as a member of two disjoint classes.
- All cardinality constraints are satisfiable.
- The transitive closure of isPartOf and hasPart properties is correctly computed.

The Protégé metrics panel confirmed: 596 total axioms, 271 logical axioms, 124 declaration axioms, 53 classes, 19 object properties, 18 data properties, and 34 individuals.

9. Ontology Graph Visualization

The following graph was generated using the OntoGraf plugin in Protégé 5.x. It visualises the complete class hierarchy and individual connections across the Transportation and Logistics Ontology.

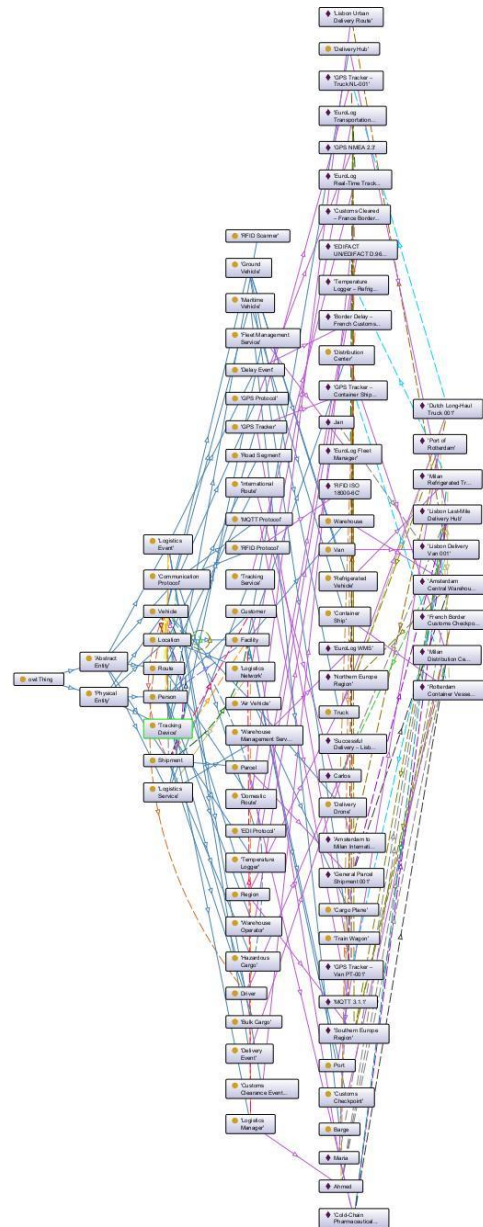


Figure 1: Onto Graf visualization of the Transportation and Logistics Ontology

10. Conclusion

This project successfully demonstrates the design and implementation of a comprehensive Transportation and Logistics Ontology in OWL 2.0. The ontology covers a realistic multi-modal logistics network spanning ground, air, and maritime transport modes, multiple facility types, cold-chain monitoring, customs procedures, and digital service integration.

The use of OWL 2 features including disjoint classes, transitive properties, inverse properties, functional properties, qualified cardinality restrictions, and existential restrictions ensures that the ontology is both expressively rich and logically sound, as confirmed by the Hermit reasoner.

The ontology provides a solid foundation for future extensions such as real-time data integration, SPARQL-based querying, and interoperability with existing logistics standards such as GS1 and UN/CEFACT.